S&P GLOBAL & CRISIL CAMPUS HACKATHON

RiskLens

Explainable News-to-Risk Engine with Wholesale-Banking Stress Testing

Transforming Unstructured Market News into Auditable Credit Shocks & Portfolio Stress Tests

- Real-Event Replay (SVB)
- Auditable Impact Formula (1-10)
- 300-Asset Wholesale Portfolio
- Reverse Stress Testing

Project Submission Details

Candidate Name: Garvit Kataria

College / Campus: Engineering Institution

Stack: Python 3.11 | ProsusAI/FinBERT | MiniLM-L6-v2 | FastAPI | Streamlit

Deterministic Offline Demo Mode | Zero API Key Dependency

Problem Statement & Solution Approach

THE PROBLEM IN WHOLESALE RISK

Information Asymmetry:
Market-moving credit shocks occur in unstructured news/social feeds hours before official rating revisions.

Sentiment Fallacy:
Generic sentiment (positive/negative) is insufficient for risk teams. 'Profits dropped 5%' and 'Bank seized by FDIC' are both negative, but have vastly different institutional consequences.

Black-Box Disconnect:
Deep neural nets output probabilities without audit trails. Risk managers cannot defend a black-box 0.89 probability to regulators or credit committees.

Static Stress Testing:
Traditional regulatory stress testing (CCAR/DFAST) is quarterly and rigid, incapable of evaluating fast-breaking geopolitical or credit events intraday.

RISKLENS SOLUTION APPROACH

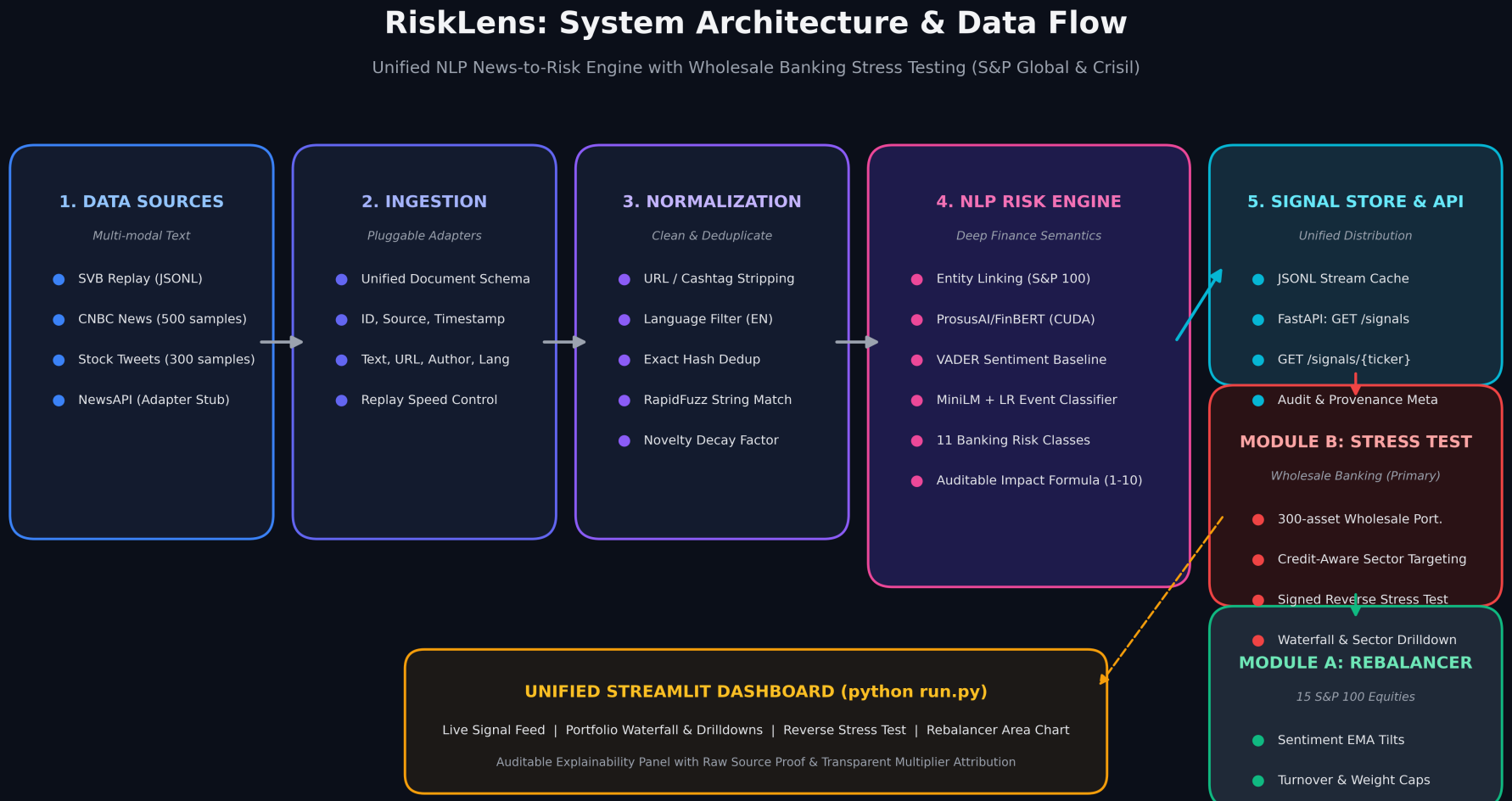
Pluggable Ingestion Pipeline:
Multi-source normalization of news and social data into a unified schema with strict deduplication and English filtering.

Domain-Aware 11-Class Taxonomy:
Maps unstructured headlines directly into Basel-aligned banking categories (Credit Event, Macro, Legal, Cyber, Supply Chain).

Auditable Impact Scoring Formula:
Transparent 1-10 formula decomposing base severity, sentiment intensity, source credibility, novelty, and cross-source corroboration.

Dynamic Wholesale Stress Testing:
Module B maps impact scores directly to spread shocks, PD multipliers, and expected loss across 300 loans, bonds, derivatives, and equities.

System Architecture & End-to-End Pipeline



ENGINEERING & MODEL CHOICES

Implementation Highlights & Defensible Technical Choices

NLP & Sentiment Engine

ProsusAI/FinBERT (fp16 on CUDA):
Specialized finance domain BERT (110M params) avoids generic NLP bias.

Score Mapping:
Net Sentiment = P(Positive) - P(Negative) in [-1.0, +1.0].

VADER Baseline:
Rule-based lexicon provided for objective benchmark and CPU fallback.

Event Classification

MiniLM-L6-v2 + Logistic Regression:
384-dim semantic embeddings with fast linear classifier; 0 training GPU overhead.

Keyword Fallback:
Deterministic taxonomy pattern matcher ensures 100% coverage on novel texts.

Taxonomy:
11 credit & market risk categories aligned with institutional portfolios.

Transparent Impact Formula

Multiplicative Scoring (1 to 10):
Impact = clip(1, 10, Base_Sev * (0.5 + |Sent|) * Cred * Novelty * Corrob).

Corroboration Boost:
Tracks temporal clusters across independent sources within 48-hour windows.

Every signal carries explanation:
Full audit metadata dictionary exposed via API and dashboard.

Wholesale Stress Testing Engine

300-Position Wholesale Portfolio:
Corporate loans, bonds, IR swaps, and equities with realistic seeds.

Credit Revaluation:
Delta EL = Delta PD * LGD * EAD + Spread MTM duration-convexity approximation.

Reverse Stress Testing:
Bisection search solves for exact headline impact score breaking 2.5% loss limit.

EMPIRICAL EVALUATION

Key Results: Benchmark vs. Naive Baseline & SVB Replay

1. SENTIMENT BENCHMARK: FinBERT vs. VADER

Model / Mapping	Accuracy	Macro-F1	Role
FinBERT (Argmax)	97.2%	0.963	Academic Benchmark
FinBERT (Threshold)	87.6%	0.876	Risk Engine Shock Scaling
VADER (Lexicon)	57.1%	0.487	Rule Baseline

Negative Class Recall: FinBERT 98.7% (F1: 0.91) vs VADER 22.4% (F1: 0.28).

FinBERT captures critical credit downside where naive lexicons miss 78% of distress.
Engine uses continuous net score P(pos)-P(neg) in [-1, 1] to scale portfolio shocks smoothly.

2. EVENT CLASSIFIER: Stratified CV Benchmark

Model	Macro-F1	Credit F1	Role
MiniLM+LR (CV)	0.679	0.923	Continuous Prob
Keyword Baseline	0.679	0.923	Rule Fallback

Honest Finding: Keyword Rules Do Most of the Work (n=67)

At n=67 the embedding classifier adds no measurable gain over keyword rules; keyword rules do most of the work. Keywords were refined while reviewing these labels, so the baseline is not fully out-of-sample (0 LR overrides at 0.55 threshold).

3. LIVE REPLAY CASE STUDY: SILICON VALLEY BANK (MARCH 8-21, 2023)

Event Ingestion & Credibility: 30 curated chronological headlines & tweets replayed with independent source credibility (0.85 news, 0.55 social).

Primary Demo Trigger: March 9, 2023 (15:00 UTC): 'SVB stock plunges 60% as bank run fears intensify' triggers Credit Event at max impact 10.0.

Credit-Aware Sector Targeting: Hits Financials (-$42.1M loss, >2.7x other sectors) with 25% spillover to others, producing -$142M net loss (-3.24%).

Signed Reverse Stress Test: 15-step bisection search pinpoints portfolio 2.5% loss limit breach at Impact 8.85 ($109.7M loss).

BUSINESS & INSTITUTIONAL RELEVANCE

Wholesale Banking Domain Impact & Regulatory Alignment

Credit & Counterparty Risk

Early Warning Indicators (EWI):
Signals detect credit deterioration before rating agencies issue formal downgrade notices.

PD / Rating Migration:
Converts qualitative negative sentiment directly into domain-informed rating transitions and expected-loss surges.

Concentration Monitoring:
Immediate asset-class, sector, and counterparty exposure aggregation.

Supervisory Stress Testing Parallels

ICAAP & Supervisory Practices:
Conceptually aligned with scenario analysis practices used in ICAAP and supervisory stress testing.

Reverse Stress Testing (SR 12-7):
Empirically solves for boundary conditions where portfolio capital thresholds are breached.

Auditable Lineage:
Every stressed valuation links to the exact source headline, timestamp, and model component score.

Tactical Capital Allocation

Reusable Core Engine:
Powers both strategic stress testing (Module B) and index rebalancing (Module A) from one feed.

Constrained Rebalancing:
Rigorous 2%-15% weight boundaries, turnover caps (5%), and sentiment EMA smoothing.

Downstream API Accessibility:
Structured REST API allows any enterprise analytics team to subscribe to signals.

Limitations, Defensible Assumptions & Future Roadmap

DEFENSIBLE ASSUMPTIONS & LIMITATIONS

Synthetic Wholesale Portfolio:
300 positions are synthetically generated with fixed random seeds for demonstration. Not actual banking institution holdings.

First-Order Pricing Approximations:
Duration/convexity bond pricing, loan expected loss (dPD * LGD * EAD), and linear DV01 swap valuations are first-order proxies, not production full revaluation.

Sentiment Is Not Causality:
Headline sentiment correlation does not imply direct market causation. Signals serve as decision-support alerts, not autonomous trade execution.

Documented Severity Table:
Base severity table reflects domain assumptions rather than full econometric event-study calibration (honestly documented in docs/ASSUMPTIONS.md).

FUTURE ROADMAP & PRODUCTION EXTENSIONS

Live Ingestion Connectors:
Implement full streaming NewsAPI, Bloomberg B-PIPE, and Reuters Elektron adapters beyond the offline replay adapter.

Multi-Factor Credit Migration:
Incorporate multi-factor Merton structural models and transition matrices (e.g. Crisil rating transition tables).

Non-Linear Derivatives Pricing:
Integrate QuantLib or full Monte Carlo simulation for multi-currency cross-asset options and structured credit derivatives.

Empirical Event Study Calibration:
Run large-scale econometric calibration against historical CRSP/Compustat abnormal returns to fine-tune severity weights.